AISC5000 Foundation of Machine Learning and AI

Homework 1 Solution

Due date: 11:59 pm, Mon, Oct 5, 2026

1. We have two observations represented by the feature vectors

$$x = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}, \qquad q = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

(a) Compute $\|x\|_1$, $\|x\|_2$, and $\|x\|_\infty$. Then compute the cosine similarity

$$\cos\theta = \frac{x^\mathsf{T} q}{\|x\|_2 \, \|q\|_2}.$$

(b) Let $S = \text{span}\{q\}$. Find the orthogonal projection $\widehat{x} = \text{proj}_S(x)$ and the residual $r = x - \widehat{x}$. Verify directly that r is orthogonal to q.

Solution.

(a) The three norms are

$$\|x\|_1 = |2| + |-1| + |2| = 5, \qquad \|x\|_2 = \sqrt{2^2 + (-1)^2 + 2^2} = 3, \qquad \|x\|_\infty = 2.$$

Moreover,

$$x^\mathsf{T} q = 2(1) + (-1)(2) + 2(2) = 4, \qquad \|q\|_2 = \sqrt{1^2 + 2^2 + 2^2} = 3.$$

Hence

$$\cos\theta = \frac{4}{3 \cdot 3} = \frac{4}{9} > 0.$$

(b) The projection coefficient is

$$\frac{q^\mathsf{T} x}{q^\mathsf{T} q} = \frac{4}{9}.$$

Thus

$$\widehat{x} = \frac{4}{9} q = \begin{bmatrix} 4/9 \\ 8/9 \\ 8/9 \end{bmatrix}, \qquad r = x - \widehat{x} = \begin{bmatrix} 14/9 \\ -17/9 \\ 10/9 \end{bmatrix}.$$

By computation,

$$q^\mathsf{T} r = \frac{14 - 34 + 20}{9} = 0.$$

2. Consider a linear model $\widehat{y} = Xw$ with

$$X = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \qquad y = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, \qquad w = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}.$$

(a) Show that the columns of X are linearly independent. Compute $X^\mathsf{T}X$ and $X^\mathsf{T}y$.

(b) Use the normal equation to find the least-squares estimator $\widehat{w}$. Compute the fitted vector $\widehat{y} = X\widehat{w}$ and the residual $r = \widehat{y} - y$. Verify the orthogonality condition $X^\mathsf{T}r = 0$.

(c) Find the eigenvalues of $X^\mathsf{T}X$ and then the singular values of X.

Solution.

(a) Let c_1 and c_2 denote the two columns of X. If $ac_1 + bc_2 = 0$, the first row gives $a = 0$, and the second row then gives $b = 0$. Thus the columns are linearly independent. Direct multiplication gives

$$X^\mathsf{T}X = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}, \qquad X^\mathsf{T}y = \begin{bmatrix} 5 \\ 6 \end{bmatrix}.$$

(b) Since

$$\det(X^\mathsf{T}X) = 3 \cdot 5 - 3 \cdot 3 = 6 \neq 0,$$

the normal equation has the unique solution

$$\widehat{w} = (X^\mathsf{T}X)^{-1}X^\mathsf{T}y = \frac{1}{6}\begin{bmatrix} 5 & -3 \\ -3 & 3 \end{bmatrix}\begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 7/6 \\ 1/2 \end{bmatrix}.$$

Therefore

$$\widehat{y} = X\widehat{w} = \begin{bmatrix} 7/6 \\ 5/3 \\ 13/6 \end{bmatrix}, \qquad r = \widehat{y} - y = \begin{bmatrix} 1/6 \\ -1/3 \\ 1/6 \end{bmatrix}.$$

Finally,

$$X^\mathsf{T}r = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix}\begin{bmatrix} 1/6 \\ -1/3 \\ 1/6 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Thus the least-squares residual is orthogonal to the column space of X.

(c) The characteristic polynomial of $X^\mathsf{T}X$ is

$$\det\begin{bmatrix} 3-\lambda & 3 \\ 3 & 5-\lambda \end{bmatrix} = (3-\lambda)(5-\lambda) - 9 = \lambda^2 - 8\lambda + 6.$$

Hence

$$\lambda_1 = 4 + \sqrt{10}, \qquad \lambda_2 = 4 - \sqrt{10}.$$

Since the singular values of X are the square roots of the eigenvalues of $X^\mathsf{T}X$,

$$\sigma_1 = \sqrt{4 + \sqrt{10}}, \qquad \sigma_2 = \sqrt{4 - \sqrt{10}}.$$

3. For a one-parameter model, define

$$g(w) = we^w, \qquad L(w) = \frac{1}{2}\big(g(w) - 2\big)^2.$$

(a) Use the product rule and chain rule to compute $L'(w)$. Evaluate $L'(0)$ and perform one gradient-descent update from $w^{(0)} = 0$ with learning rate $\eta = 0.1$.

(b) Compute $L''(0)$ and the second-order Taylor polynomial of L about $w = 0$. Use it to approximate $L(0.2)$ and compare with the exact value (to four decimal places).

Solution.

(a) First,

$$g'(w) = e^w + we^w = e^w(1 + w).$$

Applying the chain rule to the squared loss gives

$$L'(w) = \big(we^w - 2\big)e^w(1 + w).$$

At $w = 0$,

$$L'(0) = (0 - 2)(1)(1) = -2.$$

The gradient-descent update is therefore

$$w^{(1)} = w^{(0)} - \eta L'(w^{(0)}) = 0 - 0.1(-2) = 0.2.$$

(b) Since

$$g''(w) = e^w(w + 2),$$

differentiating $L'(w) = (g(w) - 2)g'(w)$ gives

$$L''(w) = \big(g'(w)\big)^2 + \big(g(w) - 2\big)g''(w).$$

Thus

$$L(0) = 2, \qquad L'(0) = -2, \qquad L''(0) = 1 + (-2)(2) = -3.$$

The second-order Taylor polynomial about zero is

$$T_2(w) = 2 - 2w - \frac{3}{2}w^2.$$

Consequently,

$$T_2(0.2) = 2 - 0.4 - \frac{3}{2}(0.04) = 1.5400.$$

The exact value is

$$L(0.2) = \frac{1}{2}\big(0.2e^{0.2} - 2\big)^2 \approx 1.5413.$$

The absolute approximation error is about 0.0013.

4. Let

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}, \qquad y = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad w = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix},$$

and consider the regularized least-squares loss

$$J(w) = \frac{1}{2} \|Xw - y\|_2^2 + \frac{1}{2} \|w\|_2^2 .$$

(a) Compute $\nabla J(w)$ and the Hessian matrix $H_J(w)$.

(b) Find the critical point w^*. Use the Hessian to explain it is the unique global minimizer.

(c) Starting from $w^{(0)} = 0$, perform one gradient-descent step with $\eta = \frac{1}{6}$. Compute $J(w^{(0)})$ and $J(w^{(1)})$. Then approximate $J(w^{(1)})$ from the second-order multivariable Taylor formula of $J(w)$ at $w^{(0)}$.

Solution.

(a) Note that

$$J(w) = \frac{1}{2}(Xw - y)^\mathsf{T}(Xw - y) + \frac{1}{2} w^\mathsf{T} w.$$

By matrix differentiation we have

$$\nabla J(w) = X^\mathsf{T}(Xw - y) + w = (X^\mathsf{T}X + I)w - X^\mathsf{T}y.$$

Here,

$$X^\mathsf{T}X = \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}, \qquad X^\mathsf{T}y = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Therefore

$$\nabla J(w) = \begin{bmatrix} 3 & 3 \\ 3 & 6 \end{bmatrix} w - \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad H_J(w) = \begin{bmatrix} 3 & 3 \\ 3 & 6 \end{bmatrix}.$$

(b) At a critical point $w^* = [w_1 \ w_2]^\mathsf{T}$, we have

$$\begin{bmatrix} 3 & 3 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

from which we obtain

$$w^* = \begin{bmatrix} 1/3 \\ 0 \end{bmatrix}.$$

Note that first leading principal minor of the Hessian is $3 > 0$, and

$$\det(H_J(w)) = 3 \cdot 6 - 3 \cdot 3 = 9 > 0.$$

Thus the Hessian matrix $H_J(w)$ is positive definite for all w, which implies J is strictly convex and hence w^* is the unique global minimizer.

(c) At $w^{(0)} = 0$,

$$\nabla J(w^{(0)}) = \begin{bmatrix} -1 \\ -1 \end{bmatrix}.$$

Therefore

$$w^{(1)} = w^{(0)} - \frac{1}{6}\nabla J(w^{(0)}) = \begin{bmatrix} 1/6 \\ 1/6 \end{bmatrix}.$$

The two loss values are

$$J(w^{(0)}) = \frac{1}{2},$$

and then

$$Xw^{(1)} - y = \begin{bmatrix} -2/3 \\ 1/2 \end{bmatrix},$$

$$J(w^{(1)}) = \frac{1}{2}\left(\frac{4}{9} + \frac{1}{4}\right) + \frac{1}{2}\left(\frac{1}{36} + \frac{1}{36}\right) = \frac{3}{8}.$$

Let $\delta = w^{(1)} - w^{(0)} = [1/6,\, 1/6]^\mathsf{T}$. The second-order Taylor formula is

$$J(w^{(0)} + \delta) = J(w^{(0)}) + \nabla J(w^{(0)})^\mathsf{T}\delta + \frac{1}{2}\delta^\mathsf{T} H_J(w^{(0)})\delta.$$

Note that

$$\nabla J(w^{(0)})^\mathsf{T}\delta = -\frac{1}{3}, \qquad \delta^\mathsf{T} H_J(w^{(0)})\delta = \frac{5}{12}.$$

Therefore

$$J(w^{(1)}) = \frac{1}{2} - \frac{1}{3} + \frac{1}{2}\cdot\frac{5}{12} = \frac{3}{8},$$

which agrees with the exact value. (J is quadratic, the approximation is exact.)